



UNITED STATES SPACE FORCE

**CYBER-ENABLED
SPACE OPERATIONS**

STRATEGIC GUIDANCE



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Foreword

The United States' ability to project spacepower is inherently reliant on its ability to operate in cyberspace. Launch, command and control, communications, and ground-based sensors and space-mission data at some point must traverse cyberspace. As our doctrine, *Spacepower*, makes clear, cyberspace operations within the network dimension of space operations represent the primary linkage to all other warfighting domains. This dependence means that cyberspace superiority is critical to space superiority at the time and place necessary for the United States to prevail in conflict.

Near-peer competitors and other potential adversaries are relentlessly conducting malicious cyber activity across all sectors of the United States' infrastructure, exploiting valuable data systems during competition, and have demonstrated the ability to deny, disrupt, degrade, destroy, or manipulate vital information and networks during conflict. Assuring delivery of space effects for the United States, its allies, and its mission partners is critical to joint force operations. The United States Space Force must defeat these threats in cyberspace to preserve freedom of action in the space domain. Current cyber capabilities are insufficient in capacity and scope to effectively defend our military space capabilities from potential adversaries, and while the partnership with United States Cyber Command is critical for our success, the ultimate responsibility of ensuring military spacepower lies with the United States Space Force.

This strategy is not system-focused or platform-based, rather it is mission-focused. The USSF will transform organizationally, operationally, tactically, and functionally to ensure Guardians at all levels are trained and ready to deliver space effects through *Cyber-Enabled Space Operations*. To achieve this, Guardians must be properly organized, trained, and equipped to compete and win against any adversary at any time. Cyber Guardians must be ready to deny adversary operational advantage against our space capabilities and provide options to hold hostile forces at risk, first to deter potential adversaries from aggression and irresponsible acts in space, and to prevail in conflict if deterrence fails.

This is not simply another challenge to be resourced. Implementation of this strategy is imperative for the well-being of the Nation. I am devoted to our success and committed to empowering our Guardians to fight and win in the cyberspace domain.



B. CHANCE SALTZMAN
General, USSF
Chief of Space Operations

Introduction

Space is vital to the United States' security, prosperity, and scientific achievement. The United States Space Force's (USSF) capabilities are an indispensable component of U.S. military power. Space is not a sanctuary and space systems are potential targets during competition and conflict. Ensuring the availability of these capabilities is fundamental to advancing U.S. security and economic prosperity and ensuring military superiority across all domains.

The USSF's ability to project spacepower relies on its ability to maneuver to, from, and through cyberspace. The USSF must not only work to ensure the cyber survivability of its space systems during development but must also actively defend its critical cyber terrain across all three space segments—ground, link, and space. Ensuring the delivery of spacepower is critical to Interagency, Joint and Combined Forces, while enabling confidentiality, integrity, and availability of space capabilities for the United States, its allies, and mission partners is an operational imperative.

Secure, resilient, and defensible cyber terrain that enables unimpeded access to the space domain is critical to winning future conflicts. In response to the increased capability and demonstrated intent of potential adversaries to hold the USSF's capabilities at risk in cyberspace, this strategy provides guidance to the Service for achieving desired cyberspace conditions that enable operations in the space domain over the next five years. Through an objectives-based approach, the USSF's cyber enterprise will move with speed and purpose across three lines of efforts (LOEs):

- Develop Cyber Guardians
- Organize and Employ Combat Ready Forces
- Deliver Resilient Space Capabilities

Space mission networks represent a critical portion of the DoD Information Network (DoDIN) and defending the USSF's networks from attack in cyberspace provides military advantages to every Joint Force Commander. Through an in-depth study of cyber effects on military space capabilities, the USSF will generate a greater understanding of the threats that space systems face in cyberspace and how to ensure mission assurance. Finally, through the security and defense of space-related cyberspace, the USSF will have the ability to share experiences and lessons learned with the interagency community to ultimately enhance the DoD's combined mission to defend the common interests of the United States, allies, and partners in space.

Strategic Threat Environment

The United States faces several complex threats in cyberspace. China and Russia present the most immediate and serious threats to United States space operations. These actors' malicious behavior threatens to undermine United States foreign policy, military operations, and economic interests. Our potential adversaries have analyzed American dependencies on space and have developed doctrine, organizations, and capabilities specifically designed to contest or deny the United States access to and through the domain. Concurrently, their use and exploitation of space are expanding significantly. They consider space access and denial as critical components of their national and military strategies. Their doctrines indicate that they view space as important to modern warfare and consider the use of cyber and counterspace capabilities as a means for reducing the United States', its allies', and partners' military effectiveness and thus key for winning future wars.

Our potential adversaries view cyber, electromagnetic warfare, and information warfare as central to their theory of victory in a conflict and have organized their forces, functions, and systems to deliberately execute integrated non-kinetic operations. By degrading U.S. critical infrastructure early in a conflict, our competitors and potential adversaries hope to blind and disrupt communication across U.S. command and control systems, delay U.S. force projection, and ultimately seize battlefield advantage. Additionally, through both military and malicious cyber actors, they have been at work for several years identifying our vulnerabilities and stealing intellectual property to advance their technology and weapon systems.

Current Situation

Numerous cyber assessments and inspections have found that legacy space networks are overly complex, have limited management, and contain decades of technical debt that places these critical capabilities at risk. Additional cyber assessments without a resourced strategy to mitigate the vulnerabilities discovered, sensor the networks, and train defensive cyber forces are not useful for a new service with responsibilities to secure the space domain.

The Cyber Guardians charged with the cyber defense of space systems are transitioning from performing a myriad of specialties in information technology and communications. As the Space Force continues to build out, there is a need for faster transition of the workforce positions to cybersecurity and defense. The desired end state is being finalized for roles, responsibilities, and authorities to properly shape the cyber workforce requirements within the DoD Cyber Workforce Framework (DCWF) to gain the required training and experience to operate at the level necessary to face our advanced cyber competitors and potential adversaries.

The USSF will employ multiple methods to field cybersecurity and cyber defense capabilities to protect vital space systems including commercial tools; development, security, operations (DevSecOps) approach; and a cyber defense platform so operators can interface with integrated capabilities. Defensive capabilities are planned for the prioritized set of critical cyber terrain. Without the requisite cyber expertise in place in the USSF cyber squadrons, cyber defense deployment has slowed down, and configurations are sub-optimal.

Fortunately, since 2020 the USSF has been building out its “Cyber Operations Roadmap.” The USSF has aligned Cyber Squadrons with Mission Deltas to provide persistent cyber defense and mission assurance capabilities to secure, monitor, and defend USSF missions that operate in, through, and from cyberspace tailored to and focused on each of the delta’s specific mission sets. Additionally, Space Operations Command (SpOC) and Space Delta 6 (DEL 6) have established the Space Cyber Center to provide the USSF with Cybersecurity Service Provider (CSSP) services, threat correlations, and cyber operations coordination.

Objectives

Expediting Integrated Operations (FY 23 – 24)

The USSF must rapidly activate Cyber Squadrons and train and mission qualify Cyber Guardians to preserve operational advantage for our most critical space missions. To accomplish this, Cyber Squadrons must be resourced, aligned to critical space mission cyber terrain, and ready. In support of the Space Force Generation (SPAFORGEN) model, Cyber Squadrons will integrate with Intelligence and Space Squadrons to provide forces to Space Combat Units and present military spacepower to

combatant commanders. Additionally, a USSF enterprise cyber protect and defend capability must be fielded and integrated into the defended weapon system, operations battle rhythm, and business processes at a speed that matches the Cyber Squadrons' operational needs.

Operationalize Defensive Cyber on the Ground and in Space (FY 24 – 26)

The USSF must establish an enduring cyber defense posture on its critical ground, link, and space segments and supporting industrial control system infrastructure. It must build robust information sharing with other services, agencies, partners and allies. Training environments with integrated space and cyber operations must be operational and accessible to all Guardians to continue training and develop advanced skill sets required of mission-ready Cyber Guardians. Finally, a robust Cyber Aggressor capability must be fielded to provide USSF operators with a comprehensive understanding of adversarial threat capabilities, while providing a means for exercising and refining defensive tactics, techniques, and procedures (TTP). As potential adversaries mature their doctrine and TTPs, USSF recognizes the need for an integrated opposition force composed of space, cyber, and electromagnetic warfare capabilities to prepare Guardians for the more challenging environments. These activities will be key in maintaining the USSF's advantage as our competitors and potential adversaries increase their capability and activities across the competition continuum.

Project Combat Power through Cyber-Enabled Space Operations (FY26+)

This future objective seeks to provide options to Joint Force Commanders (JFC) to deny adversarial operations against friendly systems as well as provide offensive effects and hold hostile forces' assets at risk. To conduct integrated space operations, cyber operators must have the situational awareness and skills required to provide the freedom of maneuverability in the cyberspace domain for space operations while simultaneously being poised to deny this freedom to our competitors and potential adversaries when called upon.

As the USSF's cyber enterprise matures, cyber operators must become masters of integrated, full-spectrum operations. A combined mastery of Cyber Defense, Electronic Warfare (EW), and other capabilities will enable the USSF's cyber operators to better defend friendly terrain and create effects for JFCs. Better cyber defense builds resilience, while integrated, full-spectrum operations expanding attack vectors needed to hold adversary assets at risk. To that end, the USSF will seek to present a Service Component to USCYBERCOM and ultimately conduct defensive and offensive cyberspace operations.

Develop Cyber Guardians

“USSF attracts, develops, inspires, empowers, and retains individuals who are principled public servants, space-minded warfighters, and bold and collaborative problem solvers.”

-General Saltzman

The USSF is faced with competition from advanced cyber adversaries. To prevail in competition and conflict, the USSF must recruit, develop, and retain world-class cyber operators. Today, many of the USSF's Cyber Guardians work in communications squadrons providing Base Operating Support Information Technology (BOS-IT). Although important to the USSF's ability to project power from its bases, the USSF does not have the workforce to perform BOS-IT functions that can be better provided through other means (e.g., commercialization). To meet and defeat its adversaries in cyberspace, the USSF must rapidly divest the BOS-IT mission through the commercialization of network services or contract workforce, transition its Cyber Guardians to cyber squadrons or mission communications units, and invest in developing world-class cyber operators.

Culture

A defining difference between providing space support and employing credible military spacepower is a warfighting culture. Operating in a benign space domain for decades has enabled complacency where a culture of cyber awareness and cybersecurity is not prevalent. The USSF was built as a digital Service where nearly every aspect of space capabilities, plans, operations, and missions depends on the reliable and secure functioning of the USSF's networks and weapon systems. All Guardians must understand that their mission is to fight and win across the competition continuum in both the space and cyberspace domains. The defense of cyberspace is not solely the responsibility of cyber operators--all Guardians must prepare to guard against malicious cyber activities.

Every Guardian must contribute to the survivability of our space capabilities and our ability to create effects in support of JFCs. The Secretary of the Air Force continues to reinforce the need for leaders to understand disruptive technologies to manage risk in development and employment. Incorporating cybersecurity, risk management, and digital infrastructure education and training throughout a Guardian's career is critical to building the workforce needed for now and the future. These areas should be core competencies for the entire Guardian workforce to build future capabilities and continually learning.

Cyber Guardians' core identity must evolve from a combat support role to that of a space warfighter. It requires fusing full-spectrum cyber operations into the spacepower core competencies of Space Security, Combat Power Projection, Space Mobility and Logistics, Information Mobility, and Space Domain Awareness.

Recruit and Retain

The USSF will employ all available avenues to recruit and retain world-class Cyber Guardians, military and civilian. Recognizing intellectual and experiential diversity as a strategic imperative that provides key organizational and team advantages, the USSF will aggressively leverage expanded personnel authority and programs (e.g., Constructive Service Credit and University Partnership Program) to recruit

the talent required to win. Internally, the USSF will encourage the crossflow and exchange of personnel from space operations, intelligence, and force modernization specialties with cyber operations.

Retention can contribute to readiness as experience enables a greater understanding of the domain. The USSF must create a culture and environment that delivers a clear value proposition to members to encourage retention through compensation, tour stabilization, recognition, and opportunities for progression.

Train

Training underpins readiness and is the foundation upon which experience is built. Cyber Guardians (enlisted, officers, and civilians) must become masters of cyberspace with the knowledge and ability to integrate cyberpower with intelligence professionals and space operators to generate military spacepower. While most current Cyber Guardians have transferred from every other military service across 9 officer and 23 enlisted military operational specialties, the USSF has consolidated its cyber force into one officer and one enlisted cyberspace operations career field to build a strong foundation of cyberspace operations expertise. Although focused on cyberspace operations, Cyber Guardians will be educated and trained with deep technical and digital skills to perform satellite communications, electromagnetic warfare, and space network operations missions that will remain critical functions of USSF operations. Cyber Guardians assigned to a force element as part of SPAFORGEN will become and remain Combat Mission Ready (CMR).

The USSF is initially leveraging the USAF cyber schoolhouses for its initial skills training needs. The USAF and USSF, through Space Training and Readiness Command (STARCOM) will identify gaps between USAF-provided training and USSF requirements. Where necessary, STARCOM will develop and implement USSF cyber training to meet the objectives of this strategy, the STARCOM *Vision and Roadmap for USSF Military Training*, and other service strategy documents.

SpOC requires a dedicated combat training unit to focus on initial qualification training, continuation, and key advanced training for all operational Cyber Guardians. This unit forms a nexus of operational cyber training activities. The Space Staff, SpOC, STARCOM, and Space Systems Command (SSC) will work together to design, develop, field, sustain and modernize Operational Test & Training Infrastructure (OTTI) to support full spectrum space operations challenging the various integrated Guardian that create the combat effects.

Educate

Cyberspace knowledge is critical to the USSF's ability to execute the five spacepower competencies. This should begin with incorporating common cyberspace education standards and requirements into curricula at every level of training and professional military education as necessary to buttress the course's core learning objectives. Educational programs should cover not only basic understanding of cybersecurity best practices, but also a foundational understanding of cyberspace operations and authorities, Electromagnetic Warfare, risk management, and the strategy, policy, and legal implications of operating in and through cyberspace to deliver spacepower.

Cyber Guardians must learn the skills necessary for space warfare and the capabilities that generate spacepower. Space 200 and 300 courses will facilitate collaboration and the development of multi-discipline Cyber Guardians at the appropriate time in their professional development. STARCOM and USSF Cyberspace Operations career field managers will identify and resource opportunities such as

professional cyber certifications or courses and relevant experiences to empower Cyber Guardians to choose the best means to build depth of knowledge while maintaining rigor.

Enhance the Cyber Force

The character of war in the space domain is unique compared to warfare in other domains, necessitating unique tactics and doctrine to develop a dedicated cadre of warfighting professionals with specialized education, training, and experience to prevail in combat. To build depth in the Cyber Operations Spacepower Disciplines, STARCOM will develop a curriculum to foster a better understanding of the reliance on and interplay between cyberspace and spacepower. This will require not only creating relevant depth in Cyber Guardians, but also practical knowledge that must be incorporated across all Spacepower Disciplines.

The USSF will deliberately develop Cyber Guardians throughout their careers. This development continuum (Figure 1) will shift from time-in-grade and time-in-service measurements to recognizing demonstrated competency. The initial training phase focuses on building the foundational knowledge, skills, and abilities to achieve a Combat Mission Ready posture as part of the SPAFORGEN model. As a Guardian progresses through training and builds operational experience, they will achieve distinct proficiency levels: Basic, Senior, and Master that reflect their demonstrated level of competency.

Figure 1. USSF Cyberspace Operations Development Continuum



The Space Force requires a cadre of deliberately developed Cyber Guardians who are masters of employing cyber capabilities at the tactical level and integrating those effects across other mission areas. To that end, STARCOM will develop a Cyber Tactician's Course (CTC) tailored to the USSF's mission requirements. The CTC will develop both officer and enlisted cadres and is the first step in building tactical cyber planning experts today and a USSF Cyber Weapons Instructor Course (WIC) in the future.

Exercises such as Space Flag and Blue Skies will allow all Guardians to mission plan and fight through contested, degraded, and operationally limited space environments against representative threats. The USSF will increase exercise participation to ensure Cyber Guardians can integrate their capabilities across the breadth of space operations and that they're confronted with representative cyber threats that our potential adversaries will employ to put the USSF's missions in jeopardy.

Cyber Aggressor capabilities that can replicate a thinking adversary are needed to build resilience. By providing cyberspace forces with repeated opportunities to fight a simulated adversary, the USSF will prepare its forces, refine TTPs, conduct mission rehearsals, and demonstrate the ability to perform mission essential tasks as part of an integrated crew presented to the Joint Force. In line with the *Vision*

for the National Space Test and Training Complex (NSTTC), STARCOM will develop and field environments to provide space warfighters interconnected, scalable, and distributed physical and digital ranges to sharpen joint warfighting solutions to prevail in conflict. This effort will include the NSTTC-Cyber component to specifically address Cyber Guardian's advanced training requirements, but also provide the opportunity to plan and train multi-domain space missions to all Guardians.

Total Force

Historically, space operations organizations have leveraged Reserve Associate Units, Individual Reservists, and Air National Guard/Army National Guard Units to access tremendous experience, knowledge, and skills to augment their force. Cyber expertise is often found in industry, and many of those experts serve in the Reserves or Guard. If Reserve Component(s) or modified active-duty constructs are approved, USSF will be prepared to integrate cyber expertise via these organizations.

Organize and Employ Combat Ready Forces

“Cybersecurity is essential to the basic functioning of our economy, the operation of our critical infrastructure, the strength of our democracy and democratic institutions, the privacy of our data and communications, and our national defense...we’ve worked with our allies and partners around the world to improve our capacity to collectively defense against and respond to cyber threats from authoritarian states that go against our national interests.”

46th President of the United States, Joseph R. Biden, *2023 National Cybersecurity Strategy*

Cyberspace is relevant to all forces that operate in and through the space domain, thus the nature of cyberspace demands that cyber forces be well integrated with space forces. The USSF’s organization and structure reflect this demand for integration across operations, acquisitions, testing, training, and intelligence communities.

The USSF’s Main Cyber Force

Delta 6 (DEL 6) will prepare, present, and integrate forces to secure and defend USSF’s space systems and missions. DEL 6 will oversee cyberspace operations, coordinate operational and administrative support functions for assigned units, and direct forces at the Space Cyber Center to carry out the following three missions:

- Serve as the Cybersecurity Service Provider (CSSP) for Space Mission Systems and Platform Information Technology (PIT) within the Air Force Area of Operations under the USCYBERCOM Command and Control (C2) construct. Additional mission analysis and the activation and maturation of a USSF service component to USCYBERCOM may require a future change to this alignment and C2 structure.
- Act as a correlation cell for the USSF’s cyber enterprise
- In concert with AFCYBER, process Joint Force Headquarters-Department of Defense Information Networks (JFHQ-DODIN) orders for the USSF’s mission systems

Cyber Squadrons (CYS) established under DEL 6 and mission aligned to Space Deltas will provide integrated cyber defense in direct support of the aligned Delta and its space missions. The first seven cyber squadrons will be aligned as follows:

- 62 CYS will align to DEL 2 and defend the Space Domain Awareness mission.
- 64 CYS will align to DEL 4 and defend the Missile Warning/Tracking mission.
- 65 CYS will align to DEL 5 and defend the Space C2 mission.
- 68 CYS will align to DEL 8 and defend the Satellite Communications and Positioning, Navigation, and Timing (PNT) missions.
- 69 CYS will align to DELs 3, 6, and 9 and defend the Space Electromagnetic Warfare (SEW), Satellite Access, and Orbital Warfare (OW) missions.
- 645 CYS will align to DELs 30 and 45 and defend the Space Launch mission.
- 15 CYS will align to DEL 15 and defend the National Space Defense Center mission.

The CYSs will defend designated space missions in coordination with the supported Space Delta/CC. CYSs will support mission commanders by conducting:

- **Hardening Activities.** To mitigate adversary exploitation, mission systems will require technological upgrades or software/process updates. CYs will provide the operational commanders and applicable program offices with mission system hardening recommendations to increase system security and resilience. CYs will recommend hardening actions and report on mission system hardening recommendations to mitigate threat-specific risks to supported commanders. Hardening activities intend to reduce the attack surface, increase resilience, and enhance deterrence through the increased difficulty of system exploitation by MCAs.
- **Hunt Operations.** Intelligence driven, threat-focused hunt operations expose malicious cyber actors (MCA) and determine MCA scope and purpose within protected USSF mission systems. Hunt operations involve active reconnaissance and counter-reconnaissance of defended systems to gain and maintain contact with an adversary while continuously developing situational awareness. Where traditional cybersecurity is passive or reactive, CYs proactively search out, identify, disrupt, and defeat threats operating within USSF mission networks or mission systems.
- **Clear Operations.** Operations to target and engage MCAs to eliminate or neutralize threats from a network or mission system, and may include actions on-DODIN to interdict, contain, disrupt, or degrade MCA activity. The objective of clearing operations is eradicating identified threats from the network or mission system.
- **Continuous Assessments.** CYs will assess the effectiveness of clearing operations, hardening actions taken by local system administrators, overall system security, and resilience. The assessment objective is to replicate threat intrusion actions to ensure vulnerabilities were mitigated, negating attempts by future MCAs employing similar intrusion and exploitation TTPs.

Additionally, CYs will provide supported mission commanders with the following three capabilities:

- **Discover and Counter Infiltration (D&CI).** D&CI is detecting, illuminating, and defeating of known or unknown threats within a defended network or mission system. The CY's employment of hunt methodologies and processes reinforces organic capability to detect, discover and characterize adversarial TTPs which evade routine cybersecurity measures.
- **Cyber Threat Emulation (CTE).** CTE is the process of emulating known adversary TTPs to replicate real-world threats to train and assess CY readiness. The cyber operator's ability to detect, respond to, and recover from the malicious cyber activity will be tested continuously throughout this process. Although CTE is normally associated with Red Teams or Cyber Aggressors, CYs must continually conduct CTE to assess crew readiness and the Delta's cybersecurity posture. The primary objective of CY CTE activities is to cover the gap between mission system administrators and defender's knowledge and known adversary TTPs by emulating specific threats.
- **Threat Mitigation.** CYs provide comprehensive, threat-focused reviews of a Delta's Mission-Relevant Terrain-Cyber. Using authorized access, CYs assess mission system cybersecurity posture and processes against specific threats to identify cybersecurity vulnerabilities. The primary objective of these activities is to mitigate threats and reduce the effects of an attack on key cyberspace terrain.

Intelligence

Cyber operations are driven by intelligence. The USSF must integrate cyber intelligence into its battle rhythm and daily operations. This requires well-articulated intelligence requirements, investment, and training to conduct the mission. Cyber squadrons must integrate intelligence professionals into their planning and battle rhythm. Cyber operators require support from the National Space Intelligence Center (NSIC) and the Space Warfighting Analysis Center to understand foreign intelligence exploitation and associated space-focused cyber capabilities.

The USSF will also improve its cyber intelligence capabilities, sharing intelligence with its partners, and unifying intelligence to support cyber operations and space operations concurrently. The USSF will leverage the Intelligence Community and industry to understand cyber threats, indications and warnings, challenges, and developments across space and cyber mission areas. The National Cyber Investigative-Joint Task Force (NCI-JTF) along with commercial information sharing will integrate the Space Cyber Center, Space Information Sharing and Analysis Center (ISAC), Cybersecurity and Infrastructure Security Agency (CISA), coalition, interagency, industry, and government partners to produce a common threat picture for cleared defense contractors and space mission owners alike.

Cyber Test and Exercise

STARCOM will coordinate with SpOC and SSC to deliver realistic, threat-informed test and exercise cyber support to Delta 6, space cyber squadrons, and the Space Cyber Center. For testing, cyber test capabilities will be provided through the integrated test force construct executed by STARCOM. Additionally, STARCOM will integrate cyber into the NSTTC to influence national partners, force modernization, and test communities for threat-representative operational testing. STARCOM and SSC will organize to support cyber exercise taskings in Delta 6, leveraging cyber range capabilities through the NSTTC.

Acquisitions and Sustainment

Failure to integrate cybersecurity into our weapon systems across the entire acquisition life cycle introduces detrimental risk to weapon systems and the mission. Cyber resiliency of USSF weapon systems depends on defining effective cybersecurity requirements, identifying acceptance criteria, and establishing verification processes in contracts. To accomplish this, cyber operators and cyber intelligence professionals must be integrated into all programs to inform requirements and cyber risk mitigation strategies. SSC and Program Executive Officers (PEOs) must ensure their workforce and organizational structure provides the expertise to ensure the development, acquisition, testing, and sustainment of cyber resilient weapon systems.

SSC is organized into a Chief Information Office (CIO) office and five directorates led by Program Executive Officers (PEOs) reporting to the Assistance Secretary of the Air Force for Space Acquisition and Integration (SAF/SQ). The SSC CIO is the senior Information technology advisor to the Commander, SSC. The CIO is responsible for matters relating to the Commands' information enterprise including Cybersecurity, Data Fabric, Information Technology, Network Infrastructure, and Software. The PEO-led directorates are charged with responsibility for developing, acquiring, testing, and sustaining mission capabilities and support operations of space systems in critical mission areas: Space Sensing; Military Communication & PNT; Battle Management Command Control and Communications; Assured Access to Space; and Space Domain Awareness and Combat Power. PEOs will develop a cybersecurity program

with required manpower to support: Cybersecurity Engineering; Assessment & Authorization Process; Cybersecurity Development Testing; and Cybersecurity Operational Testing.

Cyber Componency

Guardians are masters of the use of spacepower. In accordance with Department of Defense Directive (DoDD) 5100.01, *Functions of the Department of Defense and Its Major Components*, integrating cyber and spacepower into the joint force necessitates the use of Service Components in all Combatant Commands (CCMDs), to normalize force presentation and achieve effective integration of space forces and expertise to achieve Combatant Commander (CCDR) objectives. SPACEFOR-CYBER will provide the Commander, United States Cyber Command (CDRUSCYBERCOM) a direct subordinate to integrate space effects, activities, and expertise, in accordance with CDRUSCYBERCOM priorities.

Before the activation of SPACEFOR-CYBER, the Space Staff, SpOC, STARCOM, and SSC will form an Initial Mission Analysis Team (IMAT). This team will conduct mission analysis to determine required tasks, resources, and authorities, design an objective Component Field Command (C-FLDCOM) organization serving as a USSF Service component to the CCMD; identify and recommend sourcing solutions; and coordinate with required Department of the Air Force (DAF) stakeholders. Planning will center on the missions, capabilities, resources, and authorities required to support CDRUSCYBERCOM while identifying C-FLDCOM-specific requirements and risk mitigation mechanisms to achieve cyber superiority within the space domain.

Deliver Resilient Space Capabilities

“A resilient force can withstand, fight through, and recover from attacks...our pivot to resiliency must account for on-orbit constellations, ground stations, networks, data, and mission critical support facilities...we must enhance the responsiveness of our sustainment activities through tight integration between capability development and operations....to outcompete rivals, deter aggressors, and defeat our enemies.”

CSO Note #3, Feilding Combat-Ready Forces

The USSF must deter malicious cyber activity through denial and resilience. Vertical stovepipes of cybersecurity and a program-by-program approach to cyber defense are sure to fail with today's integrated operations. To deny the adversary advantage and benefit of aggression, the USSF must increase the difficulty associated with cyber-attack on USSF missions, as well as improve our ability to rapidly identify, protect, detect, respond, and recover from mission disruptions. This will be achieved through a combination of resilient cybersecurity and defense and the integration of data and capabilities within USCYBERCOM's Joint Cyber Warfighting Architecture (JCWA). The USSF must build and field space weapon systems that are resilient, defensible, and rapidly recover during the persistent engagement. The USSF will do this by operationalizing its vulnerability management process, leveraging advanced technology and processes, and fielding defensible weapon systems with a space enterprise cyber defense capability.

To support effective deterrence efforts in line with the 2022 NDS, the USSF must enable and develop credible cyber attribution capabilities. Capabilities enabling cyber attribution—determining the source or pathway of malicious cyber activity—must be fully developed and fielded, while integrating cyber forensics data and information provided by allies and trusted commercial partners. A robust, credible, and effectively communicated capability to rapidly identify and attribute the source of malicious cyber activity may help deter competitors and potential adversaries who would only act if their identity remained unknown.

Responsive Vulnerability Management

Vulnerability management is an operational imperative. To ensure the cyber survivability of its space capabilities, investment in cyber defense technologies must accompany broader reforms in the USSF's cybersecurity posture. Cybersecurity and vulnerability management must be prioritized and decoupled from large software releases. The USSF will develop a responsive process and associated test infrastructure to rapidly identify, prioritize, test, and mitigate cyber vulnerabilities in its weapon systems and associated critical infrastructure. Vulnerability management must be threat-informed and driven by risks to the mission. SpOC, STARCOM, and SSC will develop a governance process based on their combined authorities to evaluate risk, prioritize actions and resources, and, if necessary, hold program offices accountable if they fail to comply with these time-critical fixes.

Forefront of Technological Advancement

Operational success requires space systems designed with modularity and interoperability in mind to allow rapid scaling, pivoting, and rollout where needed. The USSF must maintain its technological edge to compete and win in the space domain. The 2020 Defense Space Strategy directed an enterprise-wide change to operations, investments, and capabilities. Therefore, the USSF will pursue rapid capability through software development to support cyber-enabled space operations. Instead of weeks to months

of lead time, software development must adopt a DevSecOps model and use continuous integration and continuous delivery (CI/CD), where software is built securely, tested, validated, and deployed rapidly.

True weapon system resiliency will come from cyber operators who master "resilient operations." Guardians should train to create resiliency in ongoing space operations despite a cyber-attack by manipulating the relevant cyber terrain. The ability to dynamically reconstitute or pivot cyber operations on complex mission system networks without the adversary's awareness is paramount to preserving space missions and exploiting the adversary's TTPs and behaviors. Software-defined networks and cloud-based technologies can enhance our resilience if appropriately employed.

Additionally, observe-orient-decide-act cycles must be accelerated to counter the speed that cyber actors can exploit a mission system. To ensure the protection of critical mission systems, the ability to parse vast and disparate amounts of data, flag and investigate malicious activity, and respond to cyber threats must be performed quickly. Given the speed required to accomplish these tasks, the USSF must employ Cloud, Artificial Intelligence (AI), and Machine Learning (ML) technologies and experts to delegate a degree of automated cyber-response actions to combat adversary intrusions and move at a faster tempo than their competitors and potential adversaries.

Defensive Cyber Operations-Space (DCO-S) Enterprise Solution

DAF Task Force Cyber Secure and the resulting DAF Cyber Campaign plan emphasizes the requirement to secure and defend space mission systems. The USSF must evaluate whether to mitigate or accept risk for critical vulnerabilities in legacy space mission systems and design cyber resiliency into future space mission systems to fight through any attack. To carry out DAF directives, the USSF needs to define an enterprise cyberspace secure and defend capability and evaluate options for delivering to the warfighter. DCO-S is purpose built and enables DEL 6 and the Mission Deltas to rapidly monitor, secure, and defend space capabilities from malicious cyber actors.

The DCO-S cyber defense platform is integral to the DoD's first space-focused CSSP. DCO-S is operated by DEL 6 to see, control key cyber terrain, detect malicious activity, and direct action to rapidly respond to, isolate, and evict intruders. The USSF needs to prioritize and validate where to integrate DCO-S into legacy systems while ensuring DCO-S is included in the design and engineering solutions for new space mission systems.

For mission assurance, cybersecurity remains a shared responsibility between operational Deltas and the program office; all space mission system owners must plan for, integrate, and align with the enterprise's secure and defend capability. This eliminates the need to pursue bespoke solutions, eliminates interoperability issues, increases efficiency, and maximizes the operational effectiveness of USSF's cyber defenders.

The USSF's current on-orbit spacecraft and those built and soon to be launched pose one of the most significant challenges in implementing effective cyber defenses. Planned spacecraft must include an upgradeable cryptography and cyber defense capability that provides real-time reporting of cyber anomalies while having the ability to adapt to evolving threats rapidly. Operators and acquisition professionals need to work together with research and industry to determine how to secure and defend on-orbit spacecraft while preserving mission capabilities during hostile and hazardous events.

Mission Risk Management Framework and Analytics

For cyber defense, cybersecurity, deterrence, mission resilience, and vulnerability management to be effective, efforts must be accurately prioritized by clearly understanding the risks posed by cyber means to space missions. The USSF must move toward a quantitative mission risk management framework validated by scientific analysis and real-world data. This framework should leverage cyber threat intelligence, and model a highly detailed understanding of system dynamics, and the historical data needed to create assessments. Historical data should be collected from cyber defense exercises with threat-representative aggressors and test data from high-fidelity simulations or emulations of mission systems to measure the resiliency of system cybersecurity and the effectiveness of cyber defenses to drive the critical actions and investments necessary to reduce the risk to mission.

SSC as Cyber Capabilities Integrator

Space Systems Command will procure and integrate an enterprise Cyber protect and defend capability for threat detection and categorization of space mission systems sufficient to educate, train, and test the readiness of space forces. Current USSF CSSP capabilities are expected to be integrated into program architectures. Future space programs shall address how the enterprise Cyber protect and defend capability integrates into their program architectural designs. Program offices must comply with DoDI 5000.90, Cybersecurity for Acquisition Decision Authorities and Program Managers. The Space Staff and Milestone Decision Authorities will ensure these requirements are addressed during appropriate funding and acquisition milestones.

STARCOM Integrated Cyber Test Forces

Any weapon system that operates in cyberspace provides an attack surface for an adversary to deny, degrade, disrupt, destroy, or manipulate (D4M) information critical to military operations (regardless of the physical "air gap"). As the Integrated Test Forces (ITF) lead, STARCOM will team with the assigned operational Delta and SSC program managers to oversee the execution of iterative and continuous cyber testing throughout the entire development, acquisition, and sustainment lifecycles for space weapon systems and cyber operations systems. Integrated Cyber Test Forces will identify cyber vulnerabilities, examine attack vectors, evaluate operational cyber capabilities, and evaluate the operational mission effects in a cyber threat environment. These actions assure the suitability, effectiveness, and system survivability of the United States space weapon systems. To provide this cyber test capability to the USSF, STARCOM will train, certify, and equip USSF cyber test experts.

Conclusion

Time is short, but through the transformative actions required by this strategy, the USSF will achieve an end state where Guardians at all levels are trained and ready to deliver space effects enabled by threat-informed, integrated, full-spectrum cyberspace capabilities.

The USSF must organize, train, equip, and resource a premier cyber force, ready and prepared to win across the continuum of conflict against the most advanced cyber actors, no matter the time or place. Through diligent planning and bold action, the USSF's Cyber Guardians will be aligned and integrated into space combat squadrons, divested of the BOS-IT mission, and invested in cyber operations. The USSF will continue to be at the forefront of technological advancement, hardening space assets, and prepared to win in the cyberspace and space domains for their Nation.